Tyrosine (SCY) membrane-partitioning PMFs: estimator comparison and concentration dependence

1 Data and conventions

- **Unbiased MD.** SCY at 0.2 M (`scy`, 26 molecules) and 0.1 M (`scy-1`, 13 molecules). Three trajectories, each split into three 200 ns blocks (400–600, 600–800, 800–1000 ns): nine blocks per concentration. Each block contains 520 000 (0.2 M) or 260 000 (0.1 M) molecule-frames. Histograms use 1 Å bins of $|z|$ ($+z$ and $-z$ folded by adding counts).
- **MD water reference.** 40–50 Å from the bilayer centre.
- **Umbrella sampling (US).** Tyrosine, three 50 ns blocks, 38 windows per leaflet, WHAM with 0.25 Å bins. Each folded block is shifted so that its mean PMF over $|z| \geq 30$ Å is zero; blocks are then averaged and the SE is a block bootstrap.
- **Temperature.** $T = 303.15$ K, so $k_B T = 2.52\,\mathrm{kJ\,mol^{-1}}$.
- **Membrane reference.** Where used, every curve is shifted so that its own minimum within $|z| = 0$–20 Å is zero. For tyrosine this minimum is the interfacial well at 11.5 Å, not the bilayer centre.
- **Estimators.** A pseudocount, $-k_B T \ln(p + \varepsilon)$ with $\varepsilon = 10^{-5}$; Poisson–Gamma; Bayesian (Dirichlet); latent logarithm; and pooled counts. The first four average the logarithms of the nine block histograms; pooled counts sum the counts first and take a single logarithm (see the leucine summary for details).

2 Main findings

1. **At 0.2 M, the five estimators agree.** Every block reaches water (Table 1), so none of them has to handle an empty water reference. Their wells lie between -16.09 and $-15.40\,\mathrm{kJ\,mol^{-1}}$, and the largest difference between any two curves over 0–50 Å is $0.69\,\mathrm{kJ\,mol^{-1}}$, comparable to one SE (0.6–0.8 $\mathrm{kJ\,mol^{-1}}$).
2. **At 0.2 M, all estimators are about 2.5 $\mathrm{kJ\,mol^{-1}}$ shallower than US** ($-18.42 \pm 0.18\,\mathrm{kJ\,mol^{-1}}$). Unlike leucine, pooled counts do not improve the agreement here.
3. **At 0.1 M, the estimators disagree.** Only 3/9 blocks reach water, and the wells span -27.7 to $-18.3\,\mathrm{kJ\,mol^{-1}}$, a spread of $9.4\,\mathrm{kJ\,mol^{-1}}$. As for leucine, this spread reflects how each method treats the six blocks without water counts. Pooled counts, which need no value for $\ln 0$, are used for the concentration comparison.
4. **Inside the membrane, concentration changes the profile slightly but reproducibly.** With the membrane reference, the 0.1 M bilayer-centre barrier is 1.1–1.7 $\mathrm{kJ\,mol^{-1}}$ higher than at 0.2 M (p(offset) = 0.0007 over 0–20 Å). This result holds when whole trajectories are resampled. Leucine, valine, and isoleucine show no such difference ($\leq 0.11\,\mathrm{kJ\,mol^{-1}}$).

5. **Tryptophan shows the same shape change; phenylalanine does not.** At 0.1 M, the tryptophan bilayer-centre barrier is $1.1 \pm 0.2\,\mathrm{kJ\,mol^{-1}}$ higher ($p \leq 0.0014$ with three trajectory blocks). Phenylalanine, whose minimum is at the bilayer centre, shows at most a marginal difference (Section 6).

6. **The membrane-to-water offset does not differ significantly.** The 0.1 M well is $2.9\,\mathrm{kJ\,mol^{-1}}$ deeper, but $p(\text{offset}) = 0.08$–0.17, because the 0.1 M water reference rests on only three blocks.

7. **The 0.1 M pooled-count profile coincides with US** in both references (well -18.33 ± 2.05 vs $-18.42 \pm 0.18\,\mathrm{kJ\,mol^{-1}}$; centre barrier 16.3 vs $16.4\,\mathrm{kJ\,mol^{-1}}$ above the well). This is consistent with a small concentration effect at 0.2 M, since US approaches the dilute limit, but it does not prove it.

Table 1: Molecule-frame counts in the 40–50 Å water window for each block (T = trajectory, B = 200 ns block).

	T1/B1	T1/B2	T1/B3	T2/B1	T2/B2	T2/B3	T3/B1	T3/B2	T3/B3
0.2 M	170	1544	3688	3204	1466	1415	1054	2677	872
0.1 M	0	197	943	0	0	1437	0	0	0

3 The estimators agree at 0.2 M

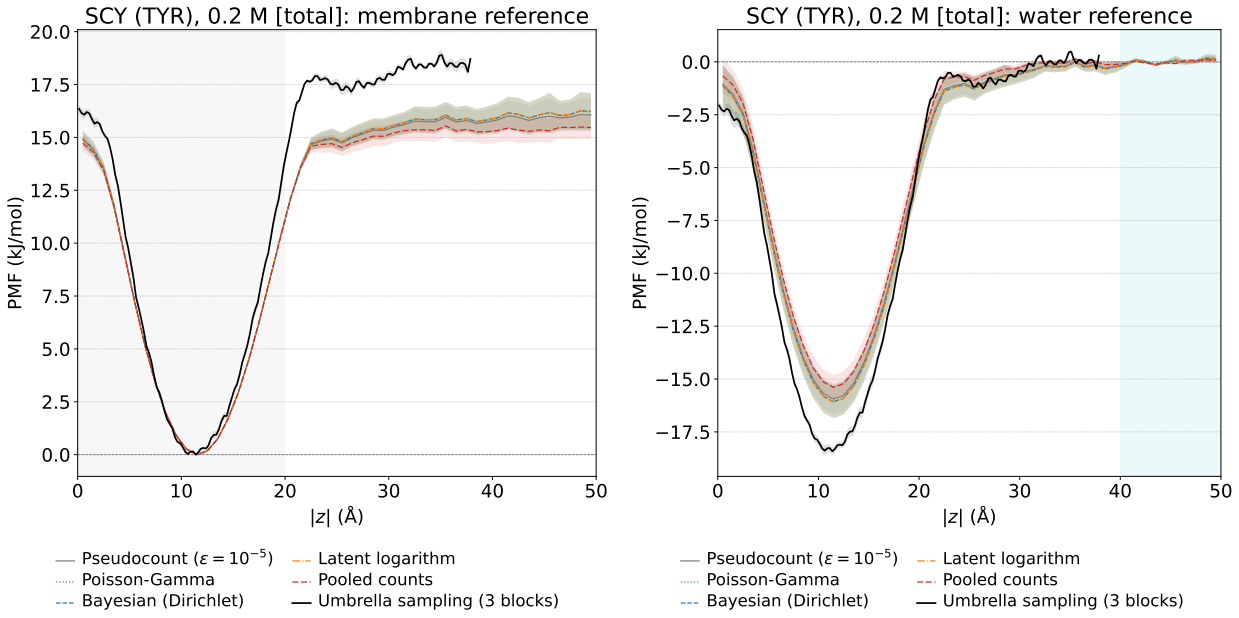


Figure 1: SCY at 0.2 M versus tyrosine US. Left: membrane reference (each minimum in 0–20 Å set to zero). Bands: untranslated SE under a fixed shift; the uncertainty of each curve's minimum is not propagated. Right: water reference (MD: 40–50 Å; US: each folded block set to zero over $|z| \geq 30$ Å). Bands: SE. The five MD curves overlap almost entirely; all are shallower than US in the well and have a lower bilayer-centre barrier relative to the well (left).

Table 2: Agreement between estimators and with US (kJ mol^{-1}). "Well": water-referenced minimum ± SE, at 11.5 Å for every curve (US: -18.42 ± 0.18 at 11.4 Å). "RMSD$_w$": root-mean-square deviation from US with the water reference over 0–38 Å (US averaged into the 1 Å MD bins). "RMSD$_m$": same with the membrane reference over 0–20 Å.

	0.2 M			0.1 M		
Estimator	Well	RMSD$_w$	RMSD$_m$	Well	RMSD$_w$	RMSD$_m$
Pseudocount	-15.95 ± 0.72	1.30	1.12	-21.69 ± 1.47	3.03	0.71
Poisson–Gamma	-16.07 ± 0.78	1.24	1.10	-27.72 ± 3.96	7.84	0.72
Bayesian	-16.08 ± 0.78	1.23	1.10	-24.51 ± 2.12	5.22	0.72
Latent	-16.09 ± 0.78	1.23	1.10	-25.82 ± 2.81	6.25	0.73
Pooled counts	-15.40 ± 0.58	1.65	1.15	-18.33 ± 2.05	0.77	0.71
Largest spread	0.69 (at 12.5 Å)			9.39 (at 11.5 Å)		

Why the estimators agree at 0.2 M. The estimators differ only in how they handle bins with zero counts in some blocks. At 0.2 M, every block visits water (Table 1) and no bin between 0 and 50 Å is empty in any block, so the pseudocount, prior, or latent model has almost nothing to fill in. The remaining spread (0.69 kJ mol^{-1}) is the difference between averaging logarithms and taking the logarithm of the pooled counts, and it is smaller than the block-to-block uncertainty.

Remaining difference with US. All 0.2 M curves are 2.3–3.0 kJ mol^{-1} shallower than US in the well. With the membrane reference, they also show a lower bilayer-centre barrier (14.7 kJ mol^{-1} above the well, versus 16.4 kJ mol^{-1} for US) and a lower water plateau (15.4–15.8 versus 18.4 kJ mol^{-1}). Because the estimators agree with each other, this difference does not come from the treatment of zero counts.

4 SCY at 0.1 M

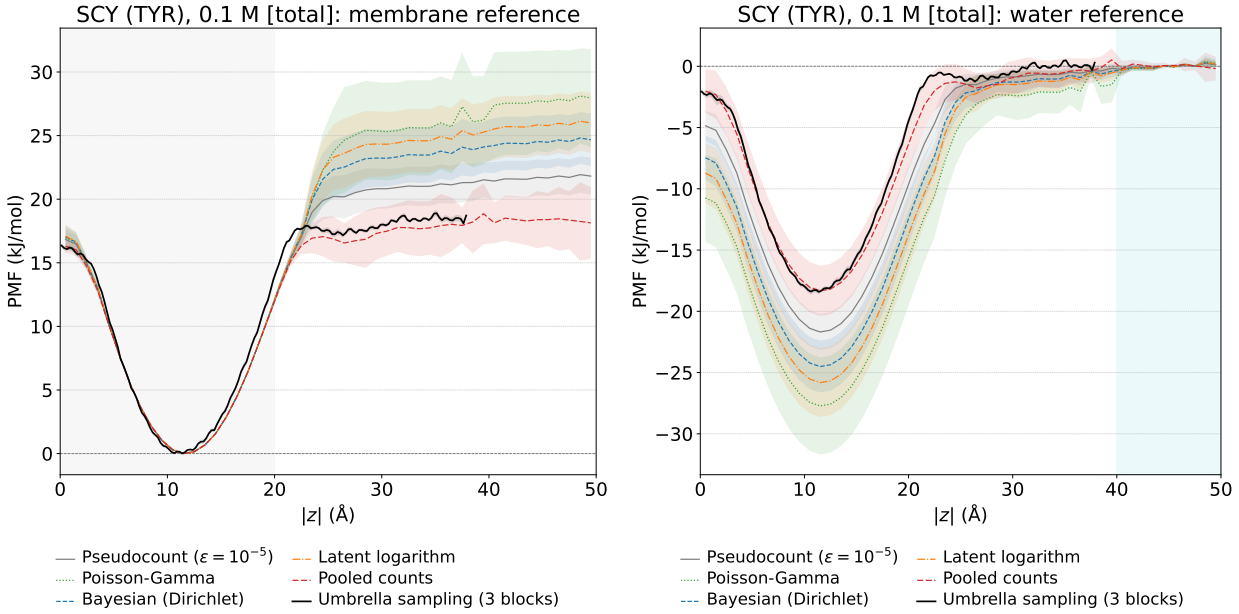


Figure 2: SCY at 0.1 M versus tyrosine US. Left: membrane reference; all estimators coincide up to about 20 Å, then reach water plateaus of 17.8–25.8 kJ mol^{-1} (US: 18.4 kJ mol^{-1}). Bands: untranslated SE under a fixed shift; the uncertainty of each curve's minimum is not propagated. Right: water reference (MD: 40–50 Å; US: each folded block set to zero over $|z| \geq 30$ Å). Only pooled counts superimpose on US. Bands: SE; the pooled-count SE excludes bootstrap replicates without water counts (97.2% of replicates are defined).

At 0.1 M, six of the nine blocks never reach the water window (Table 1). The log-averaging estimators must assign a water probability to those six blocks, and the wells then range from −21.7 (pseudocount) to −27.7 kJ mol^{-1} (Poisson–Gamma). Inside the membrane, all estimators still agree (RMSD$_m$ = 0.71–0.73 kJ mol^{-1} from US). Pooled counts give −18.33±2.05 kJ mol^{-1} and an RMSD$_w$ of 0.77 kJ mol^{-1} from US. Because the 0.1 M water reference comes from three blocks in two trajectories, its uncertainty is about 3.5 times larger than at 0.2 M (σ_a = 2.05 versus 0.57 kJ mol^{-1}).

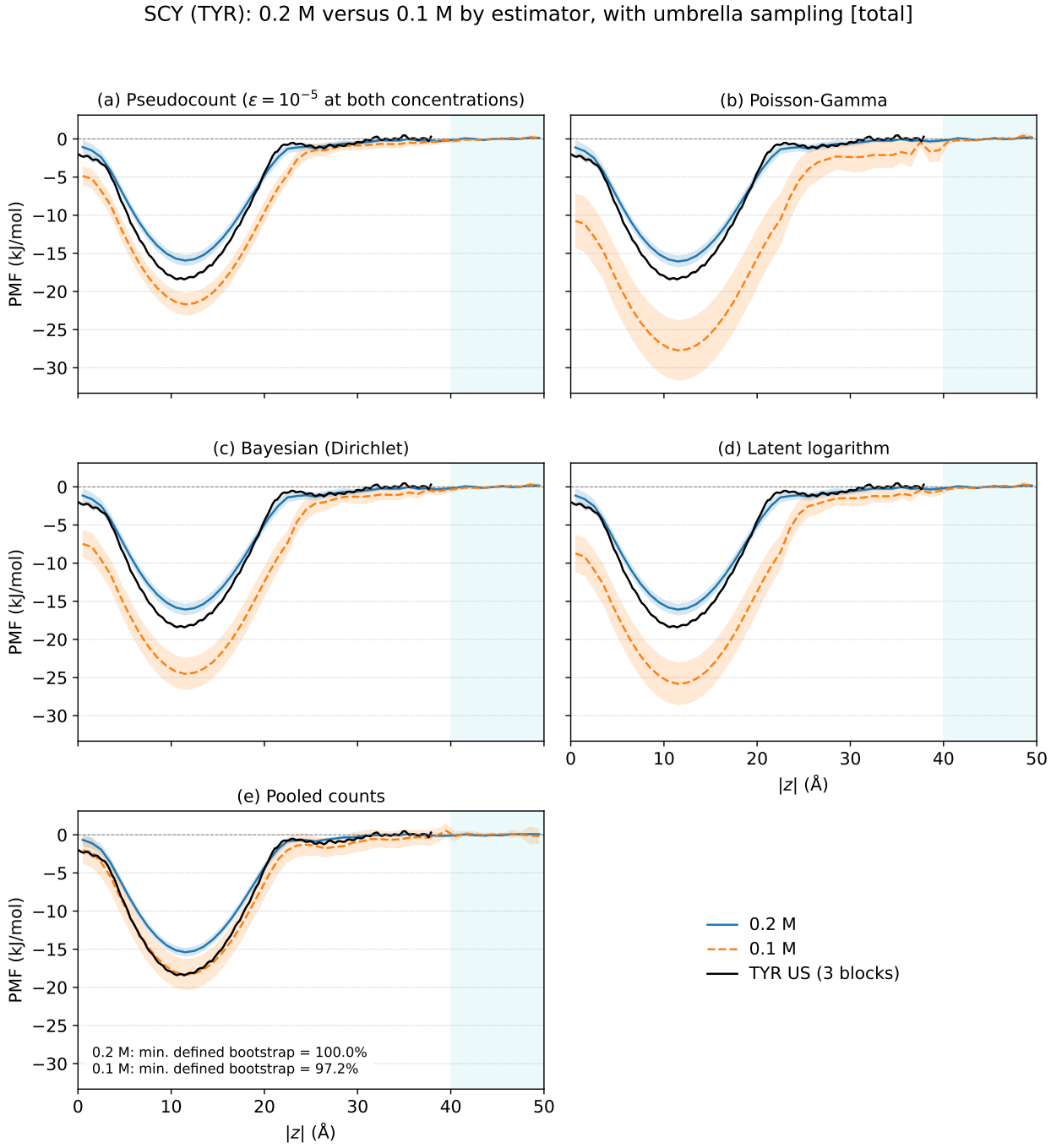


Figure 3: SCY at 0.2 M versus 0.1 M for each estimator, water reference, with tyrosine US. For every estimator, 0.1 M is deeper than 0.2 M, by 2.9 kJ mol^{-1} (pooled counts) to 11.7 kJ mol^{-1} (Poisson–Gamma). Only the pooled-count 0.1 M curve coincides with US. Water references: MD 40–50 Å; US each folded block set to zero over $|z| \geq 30$ Å. Bands: SE; the pooled-count SE excludes bootstrap replicates without water counts.

5 Is the concentration effect significant?

5.1 Test

The test uses pooled counts, as for leucine. For each concentration, the nine blocks are resampled independently 10 000 times. In each replicate, the pooled PMF and its reference are recomputed: the minimum within 0–20 Å for the membrane reference, or the mean 40–50 Å probability for the water reference. The difference $\Delta G(z) = G_{0.2\,\mathrm{M}}(z) - G_{0.1\,\mathrm{M}}(z)$ is tested over 0–20 Å and 0–50 Å with two statistics:

- $p(\max|t|)$**:** the largest studentised local difference, sensitive to a difference in any single bin;
- p(**offset**)**:** the region-averaged difference, sensitive to a systematic vertical shift.

Both use the centred bootstrap null distribution; replicates with an undefined bin are excluded. A replicate is undefined at 0.1 M when it contains none of the three water-visiting blocks, with probability $(6/9)^9 = 2.6\%$.

As a robustness check, the test was repeated with one 600 ns block per trajectory (the three 200 ns blocks of each trajectory summed), enumerating all 27×27 resample pairs. This removes any assumption that blocks from the same trajectory are independent.

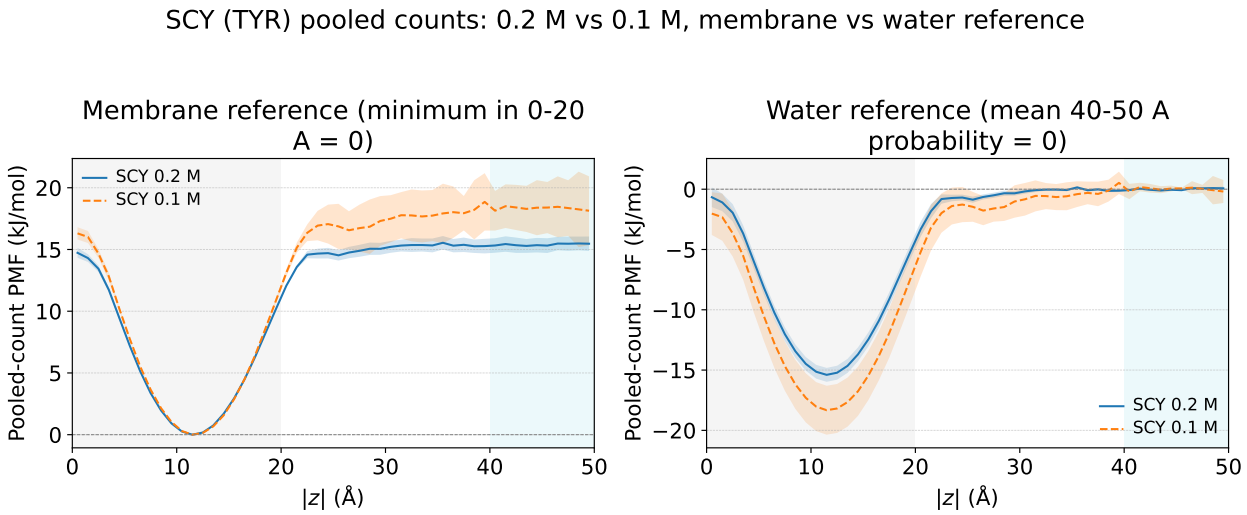


Figure 4: Pooled-count PMFs of SCY at 0.2 and 0.1 M with the membrane reference (left) and the water reference (right). Grey: 0–20 Å; cyan: 40–50 Å water window. Bands: block-bootstrap SE (defined replicates only) $\times\sqrt{9/8}$, with the reference re-estimated in every replicate. The tests of $G_{0.2\,\mathrm{M}} - G_{0.1\,\mathrm{M}}$ are given in Table 3.

Table 3: Bootstrap tests of $G_{0.2\,\mathrm{M}} - G_{0.1\,\mathrm{M}}$ for SCY with pooled counts (kJ mol^{-1}). Nine 200 ns blocks; p-values: centred bootstrap, 10 000 independent resamples per concentration, replicates with undefined bins excluded. The last column repeats p(offset) with three trajectory blocks (smallest attainable value 0.0014, reported as an upper bound).

Reference	Region	Mean ΔG	Max $\lvert\Delta G\rvert$	$p(\max\lvert t\rvert)$	p(offset)	Defined	3 traj.
Membrane	0–20 Å	-0.44 ± 0.12	1.71	0.0010	0.0007	100.0%	≤ 0.0014
Membrane	0–50 Å	-1.66 ± 0.91	3.58	0.0012	0.077	97.4%	≤ 0.0014
Water	0–20 Å	$+2.49 \pm 2.08$	3.06	0.126	0.146	97.4%	0.087
Water	0–50 Å	$+1.27 \pm 1.25$	3.06	0.093	0.165	97.4%	0.117

Interpretation.

- **A shape difference inside the membrane.** With the membrane reference, the 0–20 Å difference is significant with both statistics ($p \leq 0.001$) and with three trajectory blocks ($p \leq 0.0014$). It is concentrated at the bilayer centre: from $-1.6 \pm 0.6\,\mathrm{kJ\,mol^{-1}}$ at 0.5 Å to $-1.1 \pm 0.4\,\mathrm{kJ\,mol^{-1}}$ at 3.5 Å, so the 0.1 M central barrier is higher (16.3 vs 14.7 $\mathrm{kJ\,mol^{-1}}$ above the well). The difference is 0.4–0.7 k_BT, small but larger than for any of the aliphatic residues ($\leq 0.11\,\mathrm{kJ\,mol^{-1}}$ on average).

- **No significant change in the membrane-to-water offset.** The 0.1 M well is 2.5–2.9 $\mathrm{kJ\,mol^{-1}}$ deeper with the water reference, but $p(\mathrm{offset}) = 0.15$–$0.17$ (0.09–0.12 with three trajectory blocks). The membrane-referenced 0–50 Å test measures the same offset and gives $p = 0.077$. Its three-trajectory value ($p \leq 0.0014$) relies on SEs estimated from only three units and should not be over-interpreted.

- **Local differences.** The small $p(\max|t|)$ values reflect the bilayer centre and individual interfacial bins, where 0.1 M has few counts.

- **Agreement with US.** The 0.1 M pooled-count curve matches US in the well (difference 0.09 $\mathrm{kJ\,mol^{-1}}$), at the centre barrier, and in the water plateau, whereas the 0.2 M curves are about 2.5 $\mathrm{kJ\,mol^{-1}}$ shallower and have a lower central barrier. A weak concentration effect at 0.2 M would explain both observations, but these data do not establish it.

- **Conclusion.** Tyrosine has a reproducible concentration effect, and it is a change of shape at the bilayer centre (about 1.6 $\mathrm{kJ\,mol^{-1}}$); tryptophan shows the same effect (Section 6). The transfer free energy may also be about 3 $\mathrm{kJ\,mol^{-1}}$ more favourable at 0.1 M, in better agreement with US, but this difference is within the uncertainty of the 0.1 M water reference. More 0.1 M water-phase sampling would be needed to confirm it.

6 Phenylalanine and tryptophan

The same analysis was applied to the two other aromatic residues, phenylalanine (`scf`, `scf-1`) and tryptophan (`scw`, `scw-1`). No umbrella sampling PMF is available for them, so only the estimators and the two concentrations can be compared.

Table 4: Phenylalanine and tryptophan ($\mathrm{kJ\,mol^{-1}}$). "Water blocks": blocks with counts in the 40–50 Å window (out of 9). "Estimator wells": range of the water-referenced minima of the five estimators. "Spread": largest difference between any two estimators over 0–50 Å. "Pooled well": pooled-count minimum ± SE (at 0.5 Å for PHE and 11.5 Å for TRP).

Residue	Conc.	Water blocks	Estimator wells	Spread	Pooled well
PHE	0.2 M	7	−18.27 to −14.99	3.3	-14.99 ± 0.90
PHE	0.1 M	4	−26.66 to −16.22	10.4	-16.22 ± 4.52
TRP	0.2 M	9	−17.77 to −17.13	0.6	-17.13 ± 0.52
TRP	0.1 M	7	−23.74 to −18.94	4.8	-18.94 ± 1.86

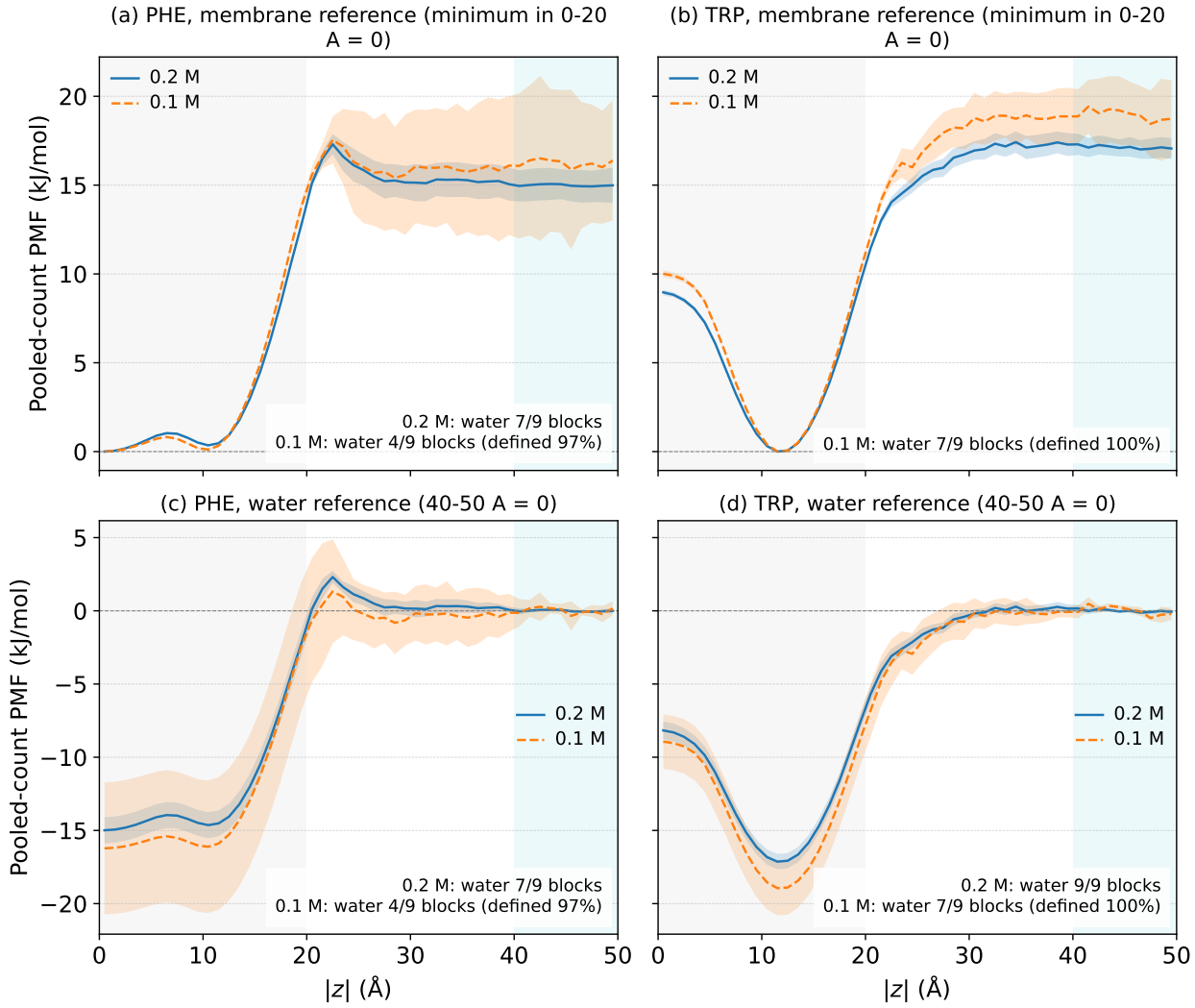


Figure 5: Pooled-count PMFs of phenylalanine (left) and tryptophan (right) at 0.2 and 0.1 M. Top: membrane reference (each minimum in 0–20 Å set to zero). Bottom: water reference (40–50 Å). Bands: block-bootstrap SE $\times\sqrt{9/8}$ (defined replicates only), with the reference re-estimated in every replicate. Notes give the number of blocks reaching water and, where below 100%, the smallest fraction of defined bootstrap replicates.

Table 5: Bootstrap tests of $G_{0.2\,\mathrm{M}} - G_{0.1\,\mathrm{M}}$ for phenylalanine and tryptophan with pooled counts ($\mathrm{kJ\,mol^{-1}}$), as in Table 3. The last column repeats p(offset) with three trajectory blocks (smallest attainable value 0.0014, reported as an upper bound).

Residue	Reference	Region	Mean ΔG	p(max $\lvert t\rvert$)	p(offset)	Defined	3 traj.
PHE	Membrane	0–20 Å	-0.12 ± 0.06	0.006	0.033	100.0%	0.10
PHE	Membrane	0–50 Å	-0.51 ± 1.94	0.006	0.69	97.4%	0.68
PHE	Water	0–20 Å	$+1.11 \pm 4.69$	0.59	0.64	99.5%	0.57
PHE	Water	0–50 Å	$+0.73 \pm 1.95$	0.40	0.52	97.4%	0.44
TRP	Membrane	0–20 Å	-0.55 ± 0.07	0.0001	0.0001	100.0%	≤ 0.0014
TRP	Membrane	0–50 Å	-1.18 ± 0.83	0.0001	0.16	99.96%	0.12
TRP	Water	0–20 Å	$+1.26 \pm 1.93$	0.35	0.46	100.0%	0.47
TRP	Water	0–50 Å	$+0.63 \pm 1.13$	0.35	0.49	99.96%	0.45

Estimators. Tryptophan behaves like tyrosine: at 0.2 M every block reaches water and the five estimators agree within $0.6\,\mathrm{kJ\,mol^{-1}}$. Phenylalanine is less well sampled in water (7/9 blocks), and its estimators already differ by $3.3\,\mathrm{kJ\,mol^{-1}}$ at 0.2 M. At 0.1 M, both residues have blocks without water counts, and the estimators diverge by 4.8 (TRP) and $10.4\,\mathrm{kJ\,mol^{-1}}$ (PHE), as for tyrosine and leucine. Pooled counts give the shallowest well in every case.

Concentration.

- **Tryptophan shows the same shape change as tyrosine.** With the membrane reference, the 0.1 M bilayer-centre barrier is $1.1 \pm 0.2\,\mathrm{kJ\,mol^{-1}}$ higher than at 0.2 M (0.5–4.5 Å), relative to the interfacial well at 11.5 Å. The 0–20 Å offset is significant with nine blocks ($p = 0.0001$) and with three trajectory blocks ($p \leq 0.0014$).
- **Phenylalanine shows only a small shape difference.** Its membrane minimum is at the bilayer centre, not at the interface. The 0–20 Å difference is $-0.12 \pm 0.06\,\mathrm{kJ\,mol^{-1}}$ on average and at most $0.9\,\mathrm{kJ\,mol^{-1}}$ near 18 Å. It is marginal with nine blocks ($p = 0.033$) and not significant with three trajectory blocks ($p = 0.10$).
- **The membrane-to-water offset does not differ for either residue.** With the water reference, the 0.1 M wells are 1.2 (PHE) and $1.8\,\mathrm{kJ\,mol^{-1}}$ (TRP) deeper, with $p = 0.46$–0.64. For phenylalanine at 0.1 M, only 4/9 blocks reach water and the well SE is $4.5\,\mathrm{kJ\,mol^{-1}}$, so even a large difference would not be detected.

Conclusion. The two residues with an interfacial well, tyrosine and tryptophan, share a reproducible concentration effect: at 0.1 M the bilayer centre is about 1–$1.6\,\mathrm{kJ\,mol^{-1}}$ less favourable relative to the well. Phenylalanine, which prefers the bilayer centre, and the aliphatic residues show no comparable effect. No residue shows a significant change in the membrane-to-water free energy, which is limited by the sampling of the water phase at 0.1 M.

7 Limitations

- Counts are molecule-frames, not independent events. Blocks within a trajectory may be correlated; the three-trajectory test addresses this but has a coarse null distribution.
- Bootstrap SEs and p-values exclude undefined replicates; at 0.1 M they are conditional on drawing at least one of the three water-visiting blocks.
- The p-values are not corrected for multiple testing (four region/reference combinations, two statistics each).

- US uses a different water window ($\geq 30\,\text{Å}$) and a single solute per leaflet, so it is not the same observable as a 0.2 M or 0.1 M ensemble. Agreement with US is supporting evidence, not proof of an unbiased estimate.